Elton Chang

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EDUCATION

Carnegie Mellon University May 2026
Master of Science in Data Analytics for Science, GPA: 3.97/4.0 Pittsburgh, PA

University of California, Santa Cruz Jun 2025
Bachelor of Science in Computer Science, GPA: 3.7/4.0 Santa Cruz, CA

EXPERIENCE

Co-Founder 2026 - Present
AMOS
- Built the Rust/Axum control layer for an AI data analyst with permission-first context, a parsed SQL subset, and statistics and chart workers outside the LLM; typed claims link to evidence for review, invalidation, and replay.
- Designed a /v1 REST API (OpenAPI, bearer auth, idempotent SQLite transactions), server-rendered review tools, and a non-root Docker image with hardened Docker Compose; GitHub Actions gates tests and benchmarks.

Founder / Technical Lead Aug 2025 - Present
Ashe System Pittsburgh, PA / San Francisco, CA
- Developed a full-stack aircraft-on-ground (AOG) risk platform: a Python/pandas scoring pipeline, FastAPI REST APIs, and a Next.js/React/TypeScript dashboard that turns operator CSV exports into explainable risk scores.
- Improved validated Precision@20 for XGBoost AOG risk ranking from 35% to 62%; tuned hyperparameters and early-warning thresholds with Optuna, increasing median warning lead time from 6.5 to 10.2 days.

Infrastructure Engineer Intern Jul 2023 - Sep 2023
Wintec Industries Inc. Newark, CA
- Reduced configuration drift across 15+ non-production hosts by provisioning test environments with Terraform.
- Cut anomaly detection time by 40% with telemetry, alerting, and dashboards for backup and replication pipelines.

Software Engineer Intern Jul 2022 - Sep 2022
Wintec Industries Inc. Newark, CA
- Built a cloud-hosted internal file management system with Java, SQL, and scheduled Python jobs that automate file ingestion, metadata indexing, and data lifecycle tracking.
- Improved onboarding and adoption by 35% by documenting its architecture and workflows for developers and users.

PROJECTS

MarketPulse AI
- Built a multi-horizon stock forecast from a weighted blend of linear-regression slope, MACD, moving-average crossovers, stochastic, and RSI, amplified by ADX and scaled by the square root of the horizon.

Enterprise AI Assistant
- Ranked organization-scoped document chunks with an 80/20 blend of OpenAI text-embedding-3-small cosine similarity and a lexical term-vector score, falling back to lexical search when embeddings are unavailable.

Zoe
- Built a pairwise ranking engine that binary-searches an insertion rank, then shifts dense 1-indexed positions in a PostgreSQL transaction under a deferred unique constraint.

Torflix
- Built an asyncio BitTorrent engine with rarest-first piece selection, endgame block duplication, peer-wire and HTTP/UDP trackers, and HTTP range responses that play video while pieces download.

OPEN SOURCE CONTRIBUTIONS

- Merged 11 pull requests into open-source projects, including Apache DataFusion (Rust SQL engine), SQLFluff (SparkSQL parser), Career Ops (JavaScript), HoloViz Panel (three UI-state fixes), and Apache Airflow.
- Merged inference-infrastructure pull requests to vLLM TPU Inference (KV-cache connector), Dynamo (VRAM-aware pytest workers), ONNX Runtime (CUDA CMake), and vLLM Production Stack (Kubernetes probe).

TECHNICAL SKILLS

Languages & Web: Python, TypeScript, JavaScript, Rust, Java, Go, C/C++, SQL, React, Next.js, Node.js, FastAPI
AI, Data & Cloud: LLM APIs, RAG, MCP, PyTorch, PostgreSQL, MongoDB, Docker, Kubernetes, Terraform, GCP